



Earn to Fire: STEMfest Newcastle info pack

InfoSec Battle Bots · 3 October 2026

What it is

Earn to Fire is a new kids' STEM mode for the IBB arena, for ages 12 to 14. Teams answer robot-themed science, maths and coding questions on an iPad, and **every right answer fires a real arena hazard**.

It launches at **STEMfest Newcastle, 6 to 8 October 2026**, for around 1,500 students in groups of about 10, roughly 50 groups a day in 7-minute slots.

How it plays

It's Red vs Blue. Each team of about five has one iPad and two bots in the arena.

1. **Two drivers** per team drive the bots, keeping their own out of the pit and pushing the other team's in.
2. **The brains** (the rest of the team) tap a weapon on the iPad and answer its question.
3. **Right answer:** that weapon fires for real in the arena, and the team scores its points.
4. **Wrong answer:** the iPad shows the right answer and explains why. That team waits 15 seconds before trying that weapon again.
5. **Recharge times are shared**, so firing a weapon also blocks the other team from it.
6. **A round lasts 2 minutes.** It ends early if both of one team's bots are pitted, which is a knockout.

Scoring

A team's score is **STEM points plus bot points**, so the brains and the drivers both win it.

Weapon	Question topic	STEM points	What fires
Flipper	Forces & levers	10	Flipper (solenoid)
The Pit	Circuits & electricity	15	Pit opens, closes after 10 s
Spinner 1	Gears & motion	20	Spinner, 8 s, random direction
Spinner 2	Code & logic	20	Spinner, 8 s, random direction
Spinner 3	Sensors & robot brains	20	Spinner, 8 s, random direction
MEGA SPIN	Boss challenge	50	All three spinners, 10 s

Bot points: +1 every second for each of your bots still in the arena, and **+50 for every enemy bot pitted.** When the clock runs out, the team with more bots still in wins the fight.

What kids learn

There are 72 KS3-level questions, 12 per weapon, all robot-themed. Examples: Ohm's law, gear ratios, binary, ultrasonic ranging, kinetic energy, and how this arena's own solenoid flipper works.

Every answer, right or wrong, comes with a one-line explanation. Teachers get a take-away sheet listing the topics, with five follow-up questions for class.

Referee tablet and safety

One tablet runs the whole game. The arena **always starts SAFE**: answers score, but nothing moves until the referee arms it.

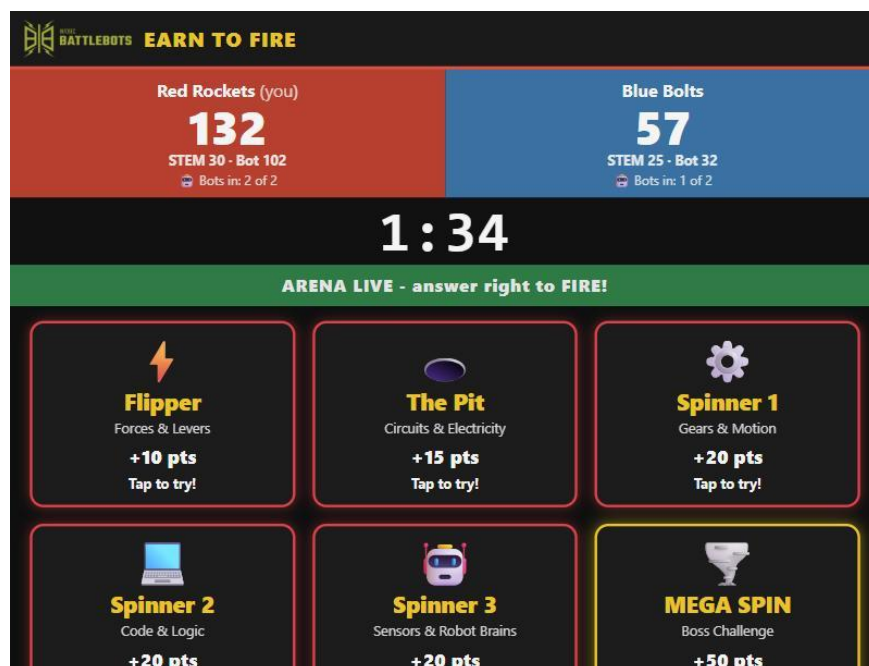
- **Round clock:** start, stop and reset. Default 2:00, with presets from 1 to 5 minutes.
- **Bot fight:** a PITTED button for each of the four bots, with undo. The kids can name their bots.
- **Standings:** STEM points, bot time, pit bonus and total for each team, updated live.
- **ALL STOP:** disarms, stops the clock, pauses the questions and switches every spinner off, at any time.
- **Auto-stops:** spinners switch off after 8 seconds (MEGA SPIN 10), and the pit closes itself after 10.

Rule for every volunteer: **disarm before any hand goes near the arena.**

Screenshots

These were taken from a demo game, so the team and bot names are made up.

The team iPad: both scores, the round clock, and the six weapons with their points.



A question: each weapon asks one on its own topic.

⚙️ **Spinner 1 - Gears & Motion - +20 pts**

A wheel is 10 cm across (diameter). How far does it roll in one turn? (use pi = 3.14)

A 10 cm

B 31.4 cm

C 314 cm

D 3.14 cm

Back

A right answer fires the weapon.

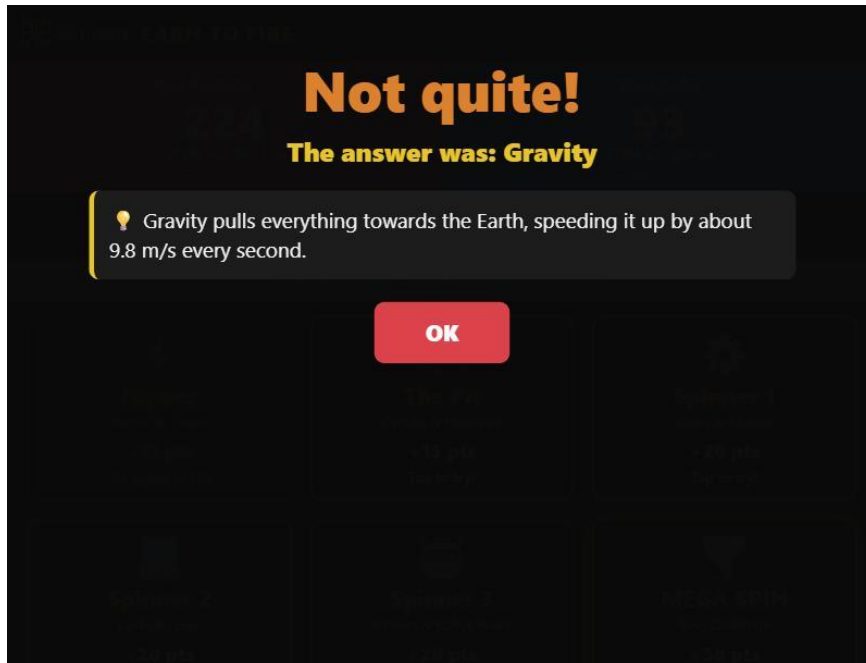
CORRECT! +20

🔥 **FIRING SPINNER 1!** 🔥

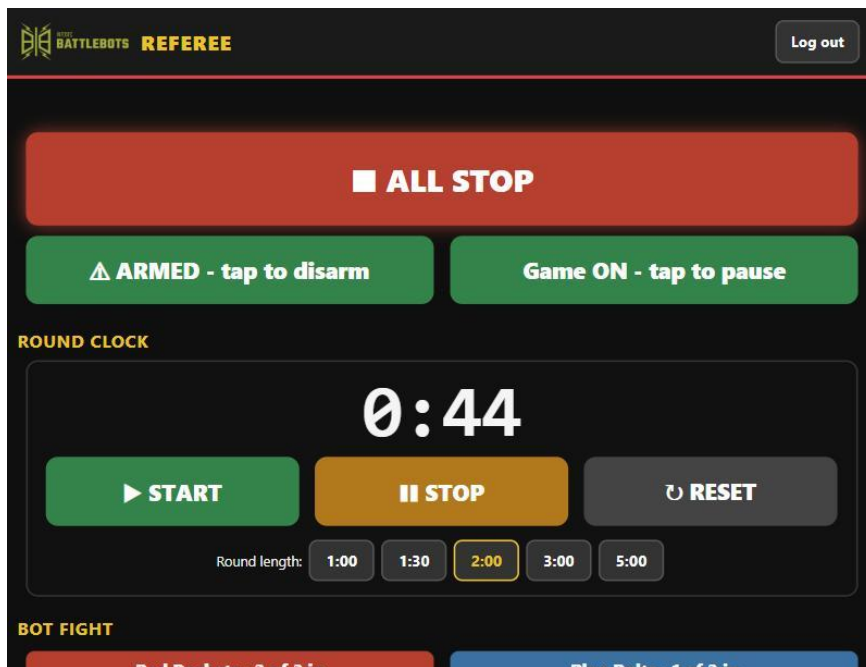
💡 Circumference = pi x diameter = 3.14 x 10 = 31.4 cm.

OK

A wrong answer explains why.



The referee tablet: ALL STOP, arming and the round clock.



Bot fight and standings: a PITTED button for each bot, and live totals.

The screenshot displays the IBB-Arena-STEM-Challenge interface. At the top, the 'BOT FIGHT' section is divided into two columns. The left column, 'Red Rockets - 2 of 2 in', features two red buttons labeled 'PITTED' with a timer 'in for 1:18' below each. The right column, 'Blue Bolts - 1 of 2 in', features a blue 'PITTED' button with a timer 'in for 1:18' and a greyed-out 'PITTED' button with a timer 'at 0:06' and an 'Undo' button below it. Below the bot fight section is the 'STANDINGS' table, which lists the teams and their performance metrics. At the bottom, the 'SCORES' section includes two input fields for 'Red Rockets' and 'Blue Bolts' with a note that the buttons adjust STEM points.

Team	STEM	Bots in	Bot time	Pit bonus	Bot pts	Fights won	TOTAL
Red Rockets	50	2	2:36	+50	206	0	256
Blue Bolts	25	1	1:24	-	84	0	109

SCORES ± buttons adjust STEM points

Red Rockets: Blue Bolts:

Status and links

- **Done:** hardware tested on 2 October: every weapon, the auto-stops and ALL STOP.
- **Next:** test the team pages in Safari on the kids' iPads.
- **Next:** print and laminate the pack: poster, briefing script, weapon cards, referee sheet, leaderboard, teacher handout.

More about the schools programme: www.infosecbattlebots.com/schools

The game is open source: github.com/Ben-PyroGuy-Doch/IBB-Arena-STEM-Challenge

Helping at STEMfest? Ask Ben for a walkthrough before Tuesday.